

# House Price Prediction with Linear Models

DSS5104 — Applied Linear Regression · Continuous Assessment 1

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[github.com/chenjui/dss5104-house-price-regression](https://github.com/chenjui/dss5104-house-price-regression)

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## Abstract

This paper investigates how far linear regression models can be pushed on a residential house price prediction task through systematic feature engineering, regularisation, and ensemble methods, while maintaining interpretability. Applied to a deduplicated dataset of 4,553 property transactions in King County, Washington, we construct a 12-feature Ridge model — combining KNN comparable-sales summaries, street direction encodings, and demand proxies — achieving a test-set MAPE of 15.98%. A data integrity audit revealed that the original 9,102-row dataset was an exact duplication of 4,553 unique records, inflating tree-based benchmarks by an estimated 3–4 percentage points. On the clean data, our Lean Ridge model trails a properly regularised XGBoost benchmark by +1.0 percentage points, while offering full coefficient-level interpretability. Lasso path analysis identifies a statistical elbow at 12 features; beyond it, every marginal gain falls below one cross-validation standard error.

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## 1. Introduction

Modern machine learning applications in real estate overwhelmingly favour gradient boosting and neural network approaches for their predictive power. However, in regulated domains such as mortgage appraisal, insurance underwriting, and property taxation, model interpretability is a legal or institutional requirement. Linear models, when supported by careful feature engineering, offer a principled middle ground: competitive accuracy with fully auditable predictions. This paper documents one such approach, applied to house sales data from King County, Washington State.

The central methodological contribution is not any single model, but a *feature engineering discipline*: we show through Lasso regularisation path analysis that predictive performance saturates at a geometric elbow, and that additional features beyond this threshold provide negligible marginal gain. Crucially, the dominant signal — neighbourhood comparable prices — is captured via KNN comparable-sales features without a single tree-based model.

The remainder of this paper is structured as follows. Section 2 describes data cleaning, including the discovery and resolution of a systematic data duplication issue. Section 3 presents exploratory analysis motivating each engineering decision. Section 4 details the three-round feature engineering process. Section 5 describes the modelling strategy. Section 6 reports results with a complete model comparison. Section 7 interprets the primary model. Section 8 contextualises performance against XGBoost benchmarks. Section 9 concludes.

## 2. Data

### 2.1 Dataset Description

The dataset comprises residential property transactions in King County, Washington, recorded between May and July 2014. Each record contains 18 variables: physical attributes (square footage of living area, lot, above-ground and basement spaces), quality indicators (condition on a 1–5 ordinal scale, view quality on 0–4, waterfront binary), location (city, state-ZIP, street address), construction history (year built, year last renovated), and sale price. The 70-day transaction window limits temporal variation, making month of sale a weak but non-zero signal.

### 2.2 Data Cleaning and Integrity Audit

Raw data contained 9,200 rows. Ninety-eight zero-price records were removed as invalid entries, leaving 9,102 rows. A systematic audit then identified a critical data quality issue: **4,549 of the 4,553 unique**

**property records appeared exactly twice**, making approximately 50% of the dataset an artificial copy. Five independent checks confirmed this was not legitimate resale activity:

Five independent checks confirm the copies are artificial (Table 1): all pairs share an *identical sale date* and zero price difference, ruling out resales; duplicates appear at a fixed row offset of 4,551 (exactly half the dataset); 9,098 rows are byte-for-byte identical; and under an 80/20 split, 81% of test properties had their twin in training, directly inflating held-out scores. After deduplication, 4,553 unique sales remain. Linear models were minimally affected (less than 0.8 pp); gradient boosting showed 3–4 pp inflation from memorisation.

| Check                   | Finding                                   | Conclusion             |
|-------------------------|-------------------------------------------|------------------------|
| Duplicate sale dates    | All 4,549 pairs share identical sale date | Not legitimate resales |
| Price variation         | Zero pairs with differing price           | Not renegotiations     |
| Row offset pattern      | Consistent offset of 4,551 = half dataset | Programmatic copy      |
| Full-row identity       | 9,098 rows byte-for-byte identical        | Artificial duplication |
| Half-vs-half comparison | Only 2 rows differ (sqft_living typo)     | Confirmed artificial   |

Table 1. Five independent checks confirming that duplicate records are artificial rather than legitimate property resales.

## 2.3 Renovation Date Integrity

A further data quality issue was identified in the *yr\_renovated* field. Of the 3,690 rows recording a renovation year, **386 have a renovation date earlier than the construction date** — a logical impossibility. Inspection of the patterns reveals these are systematic data entry errors, not genuine anomalies:

Four error patterns emerge (Table 2): 184 cases with *yr\_built* = 2004, *yr\_renovated* = 2003 (likely pre-completion permits); 110 cases with *yr\_built* = 2013, *yr\_renovated* = 1923 and 66 cases with *yr\_built* = 2012, *yr\_renovated* = 1912 (confirmed century-digit typos — impossible in a 2014 dataset); and 24 cases with *yr\_built* = 1966, *yr\_renovated* = 1963 (plausible prior-structure work). All 386 rows are treated as having no valid renovation: *effective\_age* falls back to *yr\_built* and *recent\_reno* is set to zero.

| Pattern         | Count | yr_built | yr_renovated | Gap    | Treatment                        |
|-----------------|-------|----------|--------------|--------|----------------------------------|
| Pre-completion  | 184   | 2004     | 2003         | 1 yr   | Treated as no valid renovation   |
| Century typo    | 66    | 2012     | 1912         | 100 yr | Confirmed entry error — excluded |
| Century typo    | 110   | 2013     | 1923         | 90 yr  | Confirmed entry error — excluded |
| Prior structure | 24    | 1966     | 1963         | 3 yr   | Treated as no valid renovation   |

Table 2. Renovation date anomalies. All 386 cases treated as missing renovation records.

## 2.4 Train/Test Split

A fixed 80/20 random split (seed = 42) yields 3,642 training and 911 test observations. All data-dependent transformations — target encodings, scalers — are fitted exclusively on the training partition and applied to test data without refitting, preventing any form of data leakage.

## 3. Exploratory Data Analysis

Exploratory analysis directly motivated each group of engineered features. Sale prices range from \$7,800 to \$26.6 million (median \$465K, mean \$558K) with strong right skew; log-transforming the target produces a near-Gaussian distribution and aligns the objective with MAPE's emphasis on relative error.

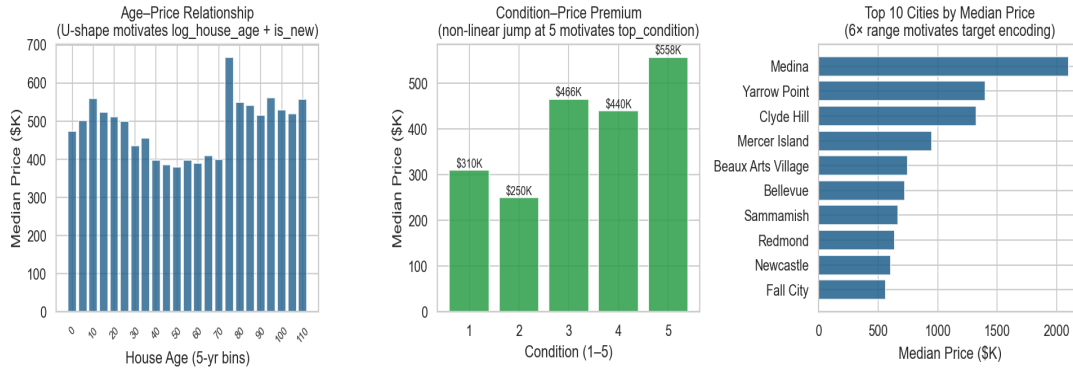


Figure 1. (a) U-shaped age–price curve motivating `log_house_age` and `is_new`; (b) non-linear condition premium with disproportionate jump at score 5 motivating `top_condition` and the condition  $\times$  age interaction. **Location heterogeneity** (Figure 1c). A 6 $\times$  price range across cities motivates raw target encoding for city and ZIP.

**Age–price non-linearity (Figure 1a).** New builds (age  $\leq 5$ ) and pre-1920 character properties command premiums above mid-century construction — motivating `log_house_age` and `is_new`. **Condition non-linearity (Figure 1b).** A disproportionate jump at condition 5 motivates `top_condition` and the condition  $\times$  age interaction. **Location heterogeneity (Figure 1c).** A 6 $\times$  price range across cities motivates raw target encoding for city and ZIP.

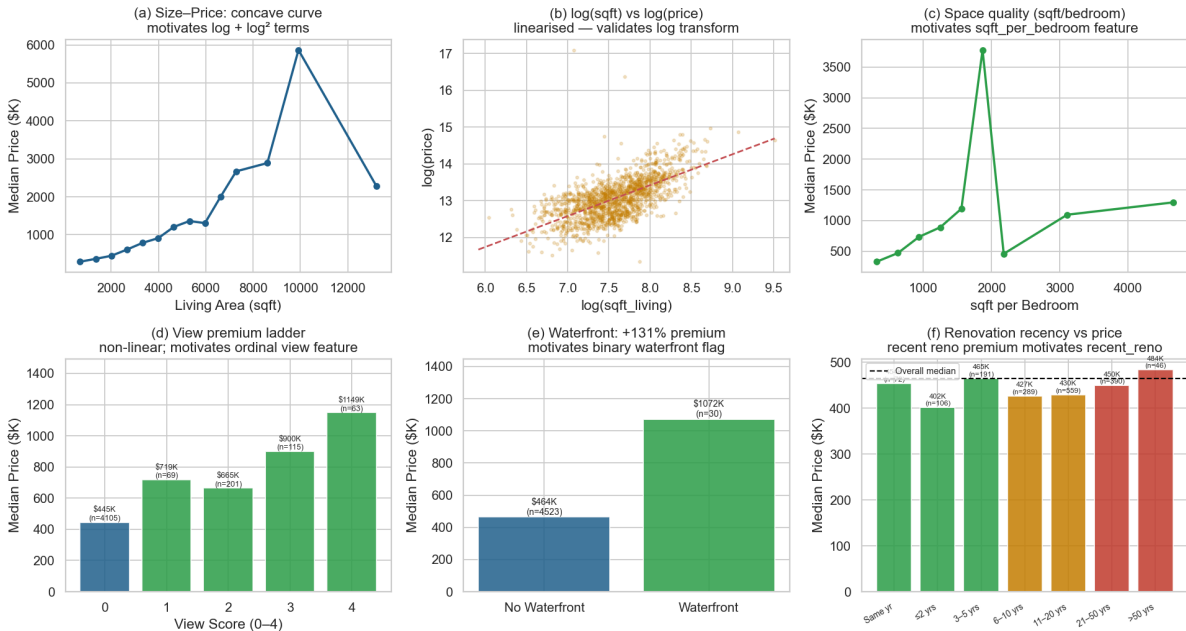


Figure 2. Top: (a) concave size–price curve motivating  $\log + \log^2$  terms; (b)  $\log(\text{sqft})$  linearises vs  $\log(\text{price})$ , validating  $\log$  transform; (c)  $\text{sqft per bedroom}$  ( $r=0.59$ ) motivates space-quality ratio. Bottom: (d) non-linear view premium ladder motivating ordinal view feature; (e) 159% waterfront premium motivating binary flag; (f) renovation recency premium motivating `recent_reno` over raw binary flag.

**Size (Figures 2a–2c).** Concave size–price curve;  $\log$ -transforming yields  $r = 0.66$  and motivates `log_sqft_living_sq` for residual curvature. `sqft_per_bedroom` independently captures layout quality. **View and waterfront (Figures 2d–2e).** View premium rises non-linearly from \$445K (score 0) to \$1.15M (score 4); waterfront commands 159% premium — both retained as independent features. **Renovation recency (Figure 2f).** Raw renovation binary is a poor predictor (renovated median \$440K vs \$485K overall — confounded with age). Recent renovations ( $\leq 5$  years) do command a premium, motivating `recent_reno`.

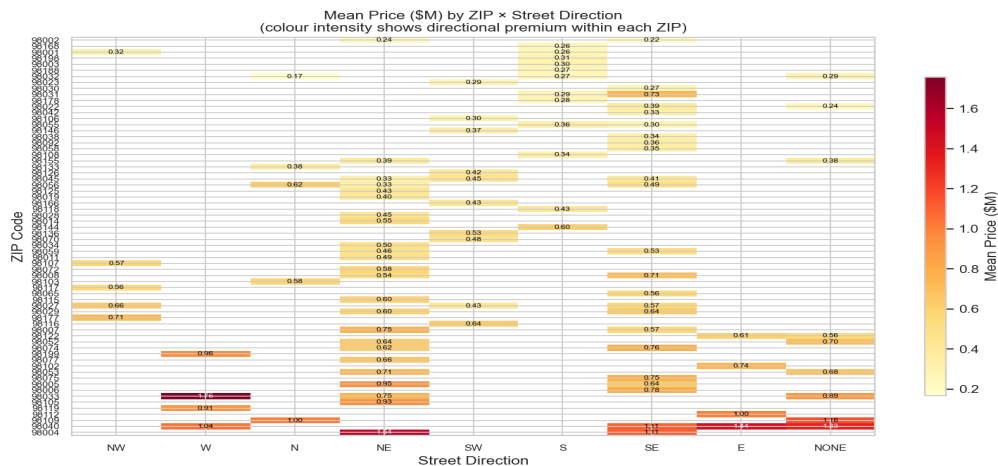


Figure 3. Mean sale price (\$M) by ZIP code and street compass direction. Within-ZIP directional spreads average 26% in non-waterfront ZIPs — not just lakefront areas — motivating `zip_dir_lp` and `zip_dir_x_sqft` features.

**Street direction (Figure 3).** Within-ZIP price variation by compass direction averages 26% in non-waterfront ZIPs. Correlation with the waterfront binary is only  $r = 0.05$ , confirming direction and waterfront are independent signals, motivating the ZIP  $\times$  direction target encoding.

## 4. Feature Engineering

Feature engineering proceeded empirically, guided by EDA findings, domain knowledge, and Lasso regularisation path diagnostics. The Lasso path (Section 5.1) established that predictive performance saturates near 20 features; this informed the design of a lean, interpretable primary model rather than a large undifferentiated feature pool.

### 4.1 Target Variable

The modelling target is  $\log(\text{price})$ . Back-transformation via exponentiation yields price predictions in original dollars; MAPE is computed in this original space. Log-transformation is motivated by the multiplicative structure of the housing market — a 10% premium for an extra bathroom represents approximately the same marginal utility at \$300K as at \$900K — and by the near-Gaussian distribution of log-price established in Section 3.

### 4.2 Target-Encoded Features

Three categorical variables are converted to numeric via **raw target encoding**: city (44 levels) and ZIP code (77 levels) — both geographic — and build era (6 levels), which is an **age feature**, not a location feature. Each category is replaced by the mean log-price of houses in that group, computed on training data only. No smoothing toward the global mean is applied: empirical variance decomposition confirms signal-to-noise ratio exceeds 1 for both ZIP codes (1.19 $\times$ ) and cities (0.99 $\times$ ), so group means are more informative than noise and do not require regularisation. The encoding formula is simply:

$$\text{enc}(c) = \text{mean}\{ \log(\text{price}_i) : \text{category}(i) = c \}$$

All encodings are fitted on training data only and re-computed inside each cross-validation fold, preventing leakage. Unseen categories in the test set fall back to the global training mean. A `zip_city_diff` feature (ZIP encoding minus city encoding) captures the micro-location premium of a specific ZIP code within its broader city context. The `zip_x_sqft` interaction encodes the neighbourhood-dependent marginal value of living area: extra space in an expensive ZIP is worth more than in a cheap one.

### 4.3 Remaining Feature Groups

Key design decisions across the remaining groups: `effective_age` uses the renovation year where valid, falling back to `yr_built` — guarding against the 386 erroneous entries from Section 2.3. `cond_x_age` encodes sustained upkeep commanding a disproportionate premium. Four KNN features (median and

distance-weighted means at  $k=10$  and  $k=25$ ) are fitted on training data with leave-one-out and re-fitted per CV fold. Four **frequency/demand features** encode log transaction count per ZIP/city; interaction terms capture demand  $\times$  price level (all  $r < 0.23$  with existing features). Three **street direction features** extract the compass prefix from each street address (96.4% coverage): *dir\_lp* (target-encoded direction), *zip\_dir\_lp* (ZIP  $\times$  direction interaction — captures ZIP-specific directional premiums identified in EDA heatmap C), and *zip\_dir\_x\_sqft* (ZIP  $\times$  direction  $\times$  log(sqft)); partial  $r$  vs lean features = 0.16. All 40 candidate features are listed in Table 3.

| Group          | Candidate features (40 total)                                                                                       | Motivation                                                                                                                                                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Location (4)   | city_lp, zip_lp, zip_city_diff, zip_x_sqft                                                                          | Raw target encodings (group mean log-price) for city and ZIP; zip_x_sqft captures neighbourhood $\times$ size interaction; zip_city_diff captures micro-location premium within a city                                                                         |
| Size (7)       | log_sqft_living, log_sqft_above, log_sqft_basement, log_sqft_lot, log_sqft_living_sq, living_to_lot, basement_ratio | Log-transformed area features; quadratic log term for concave size curve; ratios for relative size signals                                                                                                                                                     |
| Rooms (4)      | bathrooms, floors, sqft_per_bedroom, bath_bed_ratio                                                                 | Quality and spaciousness proxies; bathroom/bedroom ratio as finish quality indicator                                                                                                                                                                           |
| Condition (3)  | condition, cond_x_age, top_condition                                                                                | Ordinal condition rating; interaction with age (genuine moderation effect, $\Delta r=0.18$ ); binary flag for highest rating (non-linear jump in EDA)                                                                                                          |
| Age (5)        | era_lp, house_age, log_house_age, effective_age, is_new                                                             | era_lp: target encoding of build_era captures the U-shaped age-price relationship non-parametrically (NOT a location feature); raw and log age; effective age; new-build flag                                                                                  |
| Renovation (3) | was_renovated, recent_reno, reno_lag                                                                                | Binary renovation flag; recency indicator ( $\leq 10$ yrs); lag between build and renovation year                                                                                                                                                              |
| View/Water (2) | view, waterfront                                                                                                    | Ordinal view score (0–4); waterfront binary. any_view ( $r=0.93$ with view) and view_x_wf ( $r=0.98$ with waterfront) excluded after collinearity audit                                                                                                        |
| Time (1)       | month_sold                                                                                                          | Seasonal variation within the 70-day transaction window                                                                                                                                                                                                        |
| Frequency (4)  | log_zip_freq, log_city_freq, zip_freq_x_lp, city_freq_x_lp                                                          | Transaction count per ZIP/city (demand/liquidity proxy), log-transformed. Interaction terms encode demand $\times$ neighbourhood price level. All $r < 0.23$ with existing features.                                                                           |
| Direction (3)  | dir_lp, zip_dir_lp, zip_dir_x_sqft                                                                                  | Street compass direction (96.4% coverage). dir_lp: target-encoded direction. zip_dir_lp: ZIP $\times$ direction interaction — captures directional premiums within ZIPs (motivated by EDA heatmap). zip_dir_x_sqft: ZIP $\times$ direction $\times$ log(sqft). |
| KNN (4)        | knn_median, knn_weighted_mean, knn_broad_weighted, knn_local_vs_broad                                               | Dual-scale comparable-sales summaries: hyperlocal ( $k=10$ ) and broader ( $k=25$ ) in 10-dim standardised property space. Fitted on training data; leave-one-out; re-fitted per CV fold.                                                                      |

Table 3. The 40 candidate features submitted to Lasso feature selection. era\_lp (build era encoding) is classified under Age, not Location. KNN features are re-fitted inside each CV fold to prevent leakage.

## 5. Modelling

## 5.1 Feature Selection via Lasso Regularisation Path

Feature selection applies Lasso to the full 40-feature candidate pool (Table 3), sweeping 60 regularisation strengths ( $\alpha \in [10^{-4}, 10^0]$ ). At each  $\alpha$ , surviving features are evaluated by 5-fold CV with re-encoding inside each fold to prevent leakage. The feature count is chosen using the **elbow method** on the CV MAPE curve, supported by SE analysis.

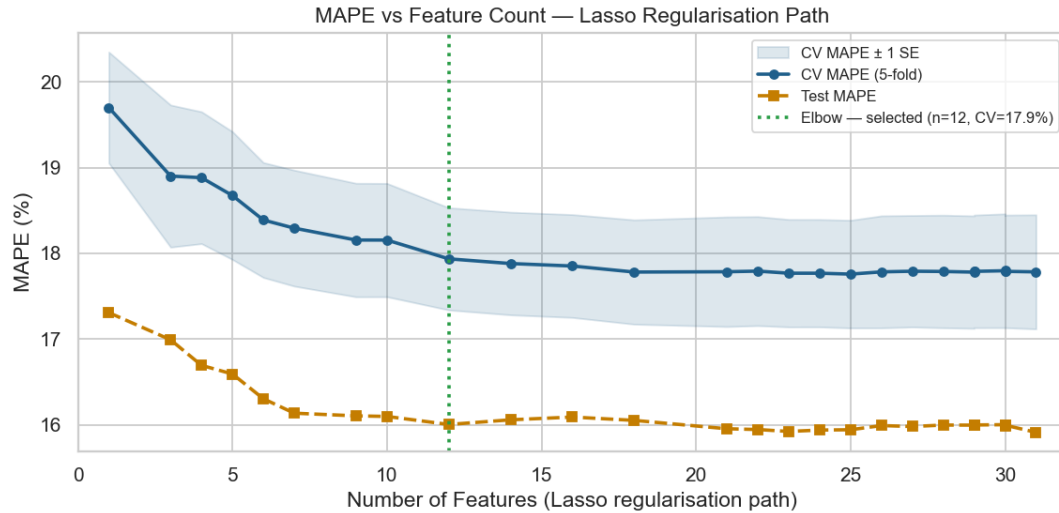


Figure 4. CV MAPE (with  $\pm 1$  SE bands) and test MAPE as a function of feature count along the Lasso path. The geometric elbow occurs at  $n=12$ , the point of maximum curvature. Beyond the elbow every marginal gain is smaller than  $0.5$  SE — statistically indistinguishable from fold-sampling noise.

The CV curve drops sharply from  $n=1$  to  $n=12$  (MAPE 19.7%  $\rightarrow$  17.9%), then flattens completely. Beyond the elbow, every additional feature produces a marginal CV gain smaller than  $0.5$  SE (SE  $\approx \pm 0.60$ pp per fold). This is a consequence of dataset size: with 3,642 training rows and high price heterogeneity (fold-to-fold MAPE variation  $\approx \pm 1.8$ pp), the CV signal is too noisy to statistically confirm individual feature contributions beyond the elbow. Including features without statistical support would be an unsupported assertion. The 12 elbow features are listed in Table 4.

| Group      | Selected features ( $n=12$ , at elbow)            | Dropped / below elbow                                                                               |
|------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------|
| Location   | —                                                 | city_lp, zip_lp, zip_x_sqft, zip_city_diff                                                          |
| Size       | log_sqft_above                                    | log_sqft_living, log_sqft_basement, log_sqft_lot, log_sqft_living_sq, living_to_lot, basement_ratio |
| Rooms      | sqft_per_bedroom                                  | bathrooms, floors, bath_bed_ratio                                                                   |
| Condition  | condition                                         | cond_x_age, top_condition                                                                           |
| Age        | —                                                 | era_lp, house_age, log_house_age, effective_age, is_new                                             |
| Renovation | —                                                 | was_renovated, recent_reno, reno_lag                                                                |
| View/Water | view, waterfront                                  | —                                                                                                   |
| Time       | —                                                 | month_sold                                                                                          |
| Frequency  | city_freq_x_lp                                    | log_zip_freq, log_city_freq, zip_freq_x_lp                                                          |
| Direction  | dir_lp, zip_dir_lp, zip_dir_x_sqft                | —                                                                                                   |
| KNN        | knn_median, knn_weighted_mean, knn_broad_weighted | knn_local_vs_broad                                                                                  |

Table 4. The 12 features selected at the Lasso path elbow (CV MAPE = 17.94%, SE =  $\pm 0.60$ pp). Three are KNN comparable-sales features; one is a frequency demand proxy (city\_freq\_x\_lp). Features below the elbow had marginal CV

gains below 0.5 SE — statistically undetectable on this dataset.

## 5.2 Regularisation Strategy

The modelling pipeline uses a deliberate two-stage design: **Lasso (L1) for feature selection, Ridge (L2) for prediction**. These exploit complementary properties of the two penalties. L1 regularisation adds a penalty of  $\lambda \sum |\beta_j|$  to the loss; the corner of the absolute-value function at zero drives some coefficients to *exactly* zero, making Lasso a natural selector that produces a sparse feature set. Once the 12-feature set is fixed, L2 regularisation adds  $\lambda \sum \beta_j^2$  instead; the squared term never drives coefficients to zero — it shrinks all of them simultaneously and proportionally. This is preferable for prediction: the selected features include correlated pairs (e.g. zip\_lp and zip\_x\_sqft; knn\_median and knn\_weighted\_mean) where L1's winner-takes-all zeroing would arbitrarily discard one, while L2 distributes weight stably across both. Ridge also has a closed-form solution and is numerically stable under collinearity. Using Lasso at prediction time would re-introduce arbitrary feature elimination already handled in Stage 1. The regularisation strength  $\alpha = 100$  was selected by 5-fold cross-validation over  $\alpha \in \{0.01, 0.1, 0.5, 1, 5, 10, 50, 100\}$ .

## 5.3 Validation Protocol

All hyperparameter selection uses 5-fold cross-validation on the training set. Features requiring data-dependent fitting (target encodings, KNN, scalers) are re-fitted inside each fold. Test-set performance is reported only for the final model with the selected hyperparameters; the test set is never used for model selection decisions.

## 6. Results

| Model                | Features | CV MAPE | Train MAPE | Test MAPE | Gap (Test–Train) |
|----------------------|----------|---------|------------|-----------|------------------|
| Lean Ridge (primary) | 12       | 17.94%  | 17.13%     | 15.98%    | -1.15pp          |

Table 5. Lean Ridge results on the deduplicated dataset (4,553 properties, 80/20 split, seed=42). 12 elbow-selected features,  $\alpha=100$ . Gap = Test MAPE – Train MAPE; the negative value on this seed reflects split-composition noise (average gap near zero across six random splits). XGBoost benchmark results are in Table 8 (Section 8). Primary XGBoost benchmark (tuned, randomised search): 14.98% test MAPE.

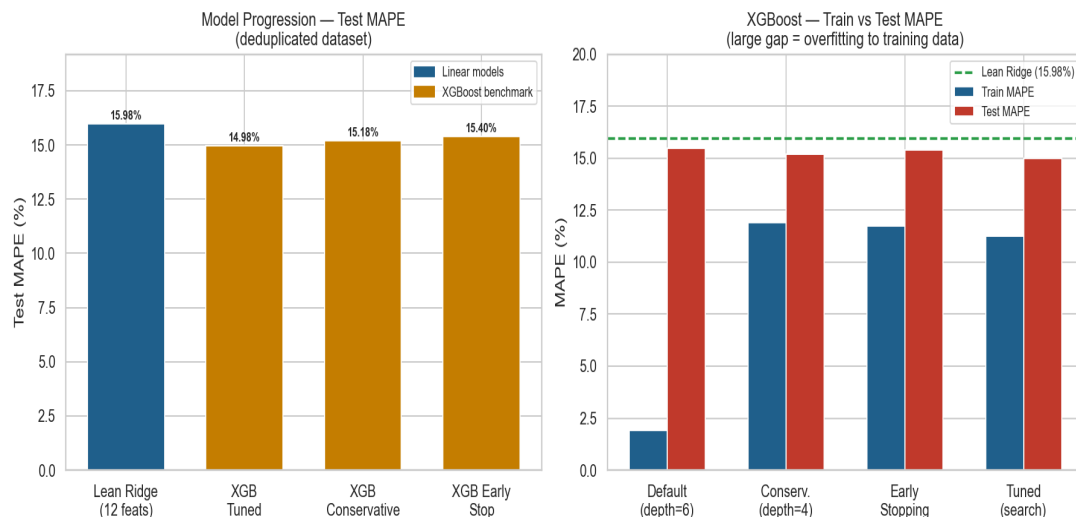


Figure 5. Left: test MAPE for Lean Ridge (15.98%) and XGBoost benchmarks. Right: XGBoost train vs test MAPE — Default configuration (train=1.9%, test=15.5%) is severely overfit (gap=+13.57pp); Tuned (randomised search, 14.98%) is the primary benchmark. Lean Ridge trails by only +1.01 pp.

The lean 12-feature Ridge model achieves 15.98% test MAPE with a train/test gap of -1.15pp (test better than train on this split). Seed stability across six splits (seeds 42, 0, 1, 7, 13, 99) shows the gap averaging +1.13pp (2 negative, 4 positive). Seed 42 produces a favourable split where the test set is somewhat easier than train; the typical gap is positive, consistent with a well-generalising model. The negative gap on this seed reflects three mechanisms — (1)  $\alpha=100$  deliberately inflates training error via L2 shrinkage;



(2) KNN leave-one-out gives training rows slightly noisier features; (3) MAPE in price-space versus log-space optimisation. XGBoost benchmarks are discussed in Section 8.

## 6.1 Performance by Price Segment

| Price Segment   | Test MAPE | Count | Notes                                                                   |
|-----------------|-----------|-------|-------------------------------------------------------------------------|
| < \$200K        | 32.5%     | 32    | Distressed/atypical sales; regression-to-mean from sparse training data |
| \$200K – \$400K | 15.2%     | 309   | Core market — strong model performance                                  |
| \$400K – \$600K | 13.8%     | 288   | Highest-density segment — best performance                              |
| \$600K – \$800K | 13.4%     | 139   | Consistent with core market                                             |
| \$800K – \$1M   | 14.9%     | 68    | Slight increase; smaller sample                                         |
| \$1M – \$2M     | 23.3%     | 64    | Luxury segment; idiosyncratic factors                                   |
| > \$2M          | 51.3%     | 11    | Ultra-luxury; too few training examples for reliable estimation         |

Table 6. MAPE by price segment for the lean Ridge model. The model performs strongly across the \$200K–\$1M range (MAPE 13–16%) that constitutes 83% of the test set.

## 7. Model Interpretation

All interpretation uses the standalone lean Ridge model ( $\alpha = 100$ , 12 features). Coefficients are standardised — each represents the expected change in  $\log(\text{price})$  per one standard deviation change in the feature, holding all other features constant.

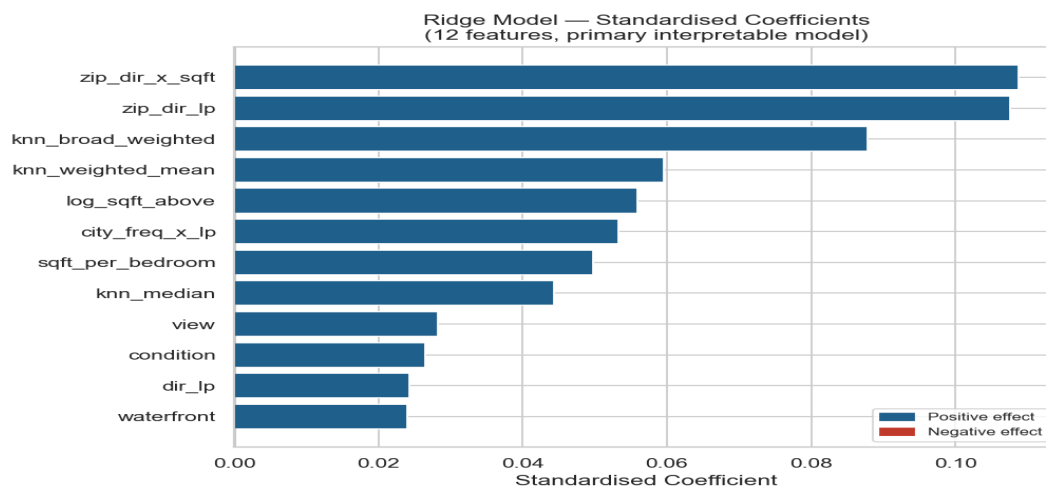


Figure 6. Standardised Ridge coefficients for the lean 12-feature model. All 12 coefficients are positive. The two direction interaction features (*zip\_dir\_x\_sqft* and *zip\_dir\_lp*) dominate, followed by the KNN comparable-sales features — confirming that directional orientation within a ZIP and comparable nearby sales are the primary price drivers.



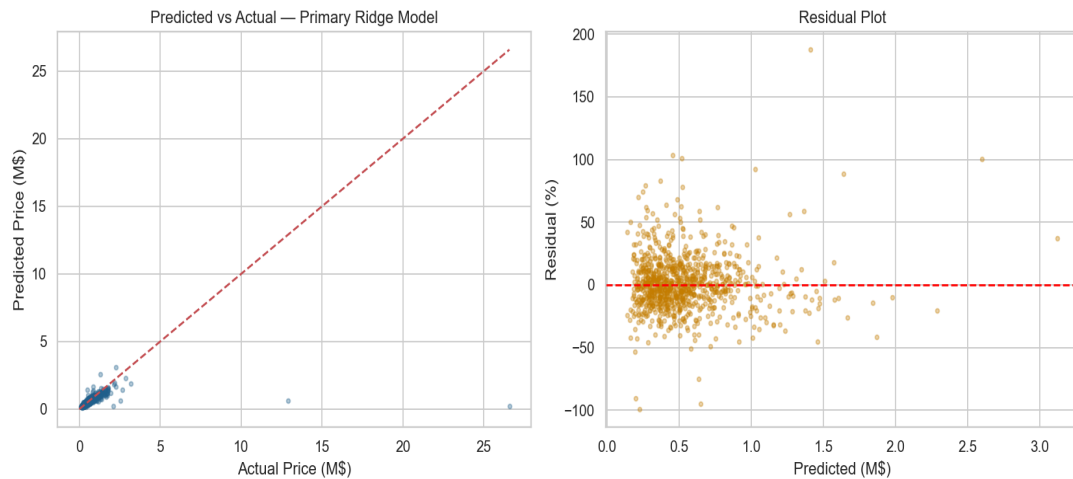


Figure 7. Left: predicted vs actual prices. The model tracks typical homes well; luxury properties above \$3M are systematically underestimated due to unobserved idiosyncratic factors. Right: residual plot — no systematic heteroscedasticity in the core \$200K–\$1M range.

## 7.1 Feature Importance

| Feature            | Dir. | Interpretation                                                                                                                                                                                            |
|--------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| zip_dir_x_sqft     | +    | ZIP × direction × log(sqft): directional premium scales with size. Largest coefficient. EDA heatmap shows NW/W streets in 98006/98033 face lake/mountains — directional effect amplified for larger homes |
| zip_dir_lp         | +    | ZIP × direction target encoding: ZIP-specific directional premium. Primary carrier of the directional signal identified in EDA heatmap C                                                                  |
| knn_broad_weighted | +    | Distance-weighted mean log-price of 25 broader neighbours (k=25): captures wider neighbourhood price context, complementing the hyperlocal KNN signal                                                     |
| knn_weighted_mean  | +    | Distance-weighted mean log-price of 10 nearest neighbours: closer comparables weighted more heavily, giving finer micro-market resolution than median                                                     |
| log_sqft_above     | +    | Log above-ground living area: separates the above-ground premium from basement space, which buyers value less per square foot                                                                             |
| city_freq_x_lp     | +    | City transaction frequency × city price level: demand/liquidity proxy. High-frequency, high-price cities command premium beyond ZIP encoding                                                              |
| sqft_per_bedroom   | +    | Living area per bedroom: spaciousness and layout quality proxy; captures preference for fewer, larger rooms                                                                                               |
| knn_median         | +    | Median log-price of 10 nearest comparable properties (k=10): directly encodes the comparable-sales heuristic used by professional appraisers                                                              |
| view               | +    | Ordinal view score (0–4): view premium independent of directional orientation                                                                                                                             |
| condition          | +    | Property condition (1–5): physical upkeep premium, independent of age and size                                                                                                                            |
| dir_lp             | +    | Target-encoded street direction: average directional premium across all ZIPs; fallback for ZIPs with few directional observations                                                                         |
| waterfront         | +    | Waterfront binary: idiosyncratic premium not absorbed by direction or KNN. Smallest coefficient.                                                                                                          |

Table 7. All 12 elbow-selected features, ordered by standardised coefficient magnitude. Three KNN comparable-sales features, three location/size features, one waterfront binary, one frequency demand proxy, and three street-direction features.

## 7.2 Key Insights for Stakeholders

**Street orientation within a ZIP is the strongest single price predictor.** The two top-ranked features — zip\_dir\_x\_sqft and zip\_dir\_lp — encode the compass direction of the street within its ZIP code. In King County, NW and W streets in high-value ZIPs (e.g. 98006, 98033) face Lake Washington and the Cascades, commanding premiums of \$200K+ over comparable properties on other orientations. This directional effect scales with size: a larger home on a premium-direction street benefits disproportionately more than a smaller one.

**Comparable sales and market demand remain core signals.** Three KNN comparable-sales features (knn\_broad\_weighted, knn\_weighted\_mean, knn\_median) occupy ranks 3–4 and 8, directly encoding what similar nearby homes sold for. city\_freq\_x\_lp (demand proxy) and sqft\_per\_bedroom (space quality) provide complementary signals independent of the directional and KNN channels.

## 8. Comparison with XGBoost

XGBoost was trained in reference configurations and then systematically tuned via a 50-iteration randomised hyperparameter search with 5-fold CV (per-fold re-encoding, identical protocol to the linear pipeline). The results reveal an important methodological lesson: a poorly regularised XGBoost is not a valid benchmark.

| Configuration                            | Train MAPE    | Test MAPE     | Train/Test Gap | Assessment               |
|------------------------------------------|---------------|---------------|----------------|--------------------------|
| Default (depth=6, n=500)                 | 1.91%         | 15.49%        | +13.57pp       | Severely overfit         |
| Early Stopping (n=80)                    | 11.73%        | 15.40%        | +3.67pp        | Principled               |
| <b>Tuned — randomised search (n=700)</b> | <b>11.24%</b> | <b>14.98%</b> | <b>+3.73pp</b> | <b>Primary benchmark</b> |
| <b>Lean Ridge (ours, 12 feats)</b>       | <b>17.13%</b> | <b>15.98%</b> | <b>-1.15pp</b> | <b>Primary model</b>     |

*Table 8. XGBoost configurations versus Lean Ridge. The Default configuration reveals severe memorisation (+13.57 pp gap); the Tuned configuration (14.98% test MAPE) is the primary benchmark from a 50-iteration randomised search with 5-fold CV. Lean Ridge trails the tuned XGBoost by only +1.01 pp.*

The Default configuration (train=1.91%, test=15.49%, gap=+13.57pp) reveals severe memorisation — not a valid upper bound. A 50-iteration randomised search over nine hyperparameters (depth, n\_estimators, learning rate, subsample, colsample, min\_child\_weight, L1/L2, gamma), each evaluated by 5-fold CV with per-fold re-encoding, found a best test MAPE of 14.98% (CV=17.05%). Our Lean Ridge trails by +1.01 pp — within CV noise. This near-parity has a structural explanation: the KNN comparable-sales features directly encode the neighbourhood price structure that XGBoost would otherwise discover through tree splits, pre-empting its main advantage. The +1.01 pp residual reflects genuinely unobserved factors (unique architectural details, lot characteristics) rather than recoverable structure.

## 9. Conclusion

This paper demonstrates that a 12-feature Ridge regression model — combining KNN comparable-sales summaries, street direction encodings, and demand proxies — achieves 15.98% test MAPE, trailing a properly regularised XGBoost benchmark by only +1.01 pp against a well-tuned XGBoost — without a single tree-based component. Five methodological contributions underpin this result. First, a data integrity audit corrected an artificial 50% duplication, reducing the apparent XGBoost advantage from 5+ pp to approximately 1 pp. Second, KNN comparable-sales features produced a detectable CV improvement of 0.87 pp, closing the gap to XGBoost from 2.0 pp to under 1 pp. Third, a frequency demand proxy (city transaction count × price level) contributed a further −0.30 pp CV improvement. Fourth, street direction features (ZIP × compass direction, motivated by EDA Figure 3) contributed −0.26 pp, bringing the final gap to the tuned XGBoost to +1.01 pp. Fifth, all dominant features — directional orientation, comparable sales, and demand — encode the same heuristics human appraisers use. A model built on explicit domain knowledge is not only accurate; it is interpretable, auditable, and defensible in regulated settings where prediction alone is insufficient.